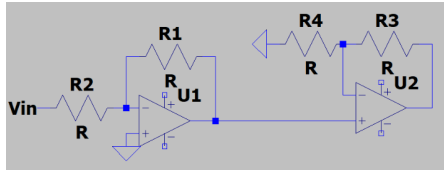


Two amplifier circuits, one an inverting amplifier with A0,1 and one non-inverting amplifier with A0,2, are cascaded to produce an overall circuit with a DC gain of 100 V/V (applicable to each subquestion). Both subcircuits use the same op amp, which has an open-loop gain of 500 V/mV and a gain bandwidth product of 15 MHz.

Using these resistor numbers



(a) If both amplifier subcircuits have the same [closed-loop] bandwidth, what must be the bandwidth of the full circuit? Further, what is the gain of each subcircuit? Hint: what is the relationship between gain and feedback factors?

$$a_0 f_b = f_t = 15\text{MHz} ; f_b = 15\text{MHz} / 500\text{k} = 30\text{Hz}$$

$$f_b = 1/B$$

$$A_{01} = -R_1/R_2$$

$$1/B_{\beta 1} = R_2/(R_1+R_2) = 1/(1 + R_1/R_2) = 1/(1 - A_{01})$$

$$A_{01} = 1 - 1/B_{\beta 1}$$

$$A_{02} = 1 + R_3/R_4$$

$$1/B_{\beta 2} = R_4/(R_3+R_4) = 1/(1 + R_3/R_4) = 1/A_{02}$$

$$A_{02} = 1/B_{\beta 2}$$

$$A_{01} \cdot A_{02} = -100 = (1/B_{\beta 2})(1 - 1/B_{\beta 1}) = (1/B - 1/B^2)$$

$$-(1/B)^2 + (1/B) + 100 = 0$$

$$(-1 \pm \sqrt{1 - 4(-1)(100)}) / 2(-1) = 0.5 \pm 0.5\sqrt{401}$$

$$1/B = 0.0951249$$

$$A_{01} = 10.51249$$

$$A_{02} = -9.51249$$

$$f_b = 0.0951249 \cdot 15\text{MHz} = 1.427\text{MHz (per circuit closed loop bandwidth)}$$

$$|A| = A_0 / \sqrt{1 + (f/f_b)^2}$$

$$|A|_{-3\text{dB}} = A_0 / \sqrt{2} = A_0 / \sqrt{1 + 1} = A_0 / \sqrt{1 + (f_b/f_b)^2} ; f = f_b$$

$$|A| = A_{01} / \sqrt{1 + (f/f_b)^2} \cdot A_{02} / \sqrt{1 + (f/f_b)^2} = A_{01} A_{02} / (1 + (f/f_b)^2)$$

$$|A|_{-3\text{dB}} = A_0 / \sqrt{2} = A_{01} A_{02} / (\sqrt{2}) = A_{01} A_{02} / (1 + \sqrt{2} - 1) = A_{01} A_{02} / (1 + (\sqrt{2} - 1)^2)$$

$$= A_{01} A_{02} / (1 + (\sqrt{2} - 1)^2) ;$$

$$f_b = \sqrt{(\sqrt{2} - 1)} f_b = 918.327\text{kHz (full circuit closed loop bandwidth)}$$

$$A_{01} = 10.51249$$

$$A_{02} = -9.51249$$

(b) If both amplifier subcircuits have the same gain magnitude, what is the bandwidth of each subcircuit and the full circuit? Hint: it should be very similar to your answer from the previous problem.

$$A_{01}A_{02} = -100 ; A_{01} = -10, A_{02} = 10$$

$$\beta_1 = 1/(1 - A_{01}) = 1/11$$

$$f_{B1} = f_t / 11 = 15\text{MHz}/11 = 1.3636\text{MHz}$$

$$\beta_2 = 1/A_{02} = 1/10$$

$$f_{B2} = f_t/10 = 1.5\text{MHz}$$

$$A = A_{01}/(1 + f/f_{B1}) * A_{02}/(1 + f/f_{B2}) = A_{01}A_{02}/((1 + f/f_{B1})(1 + f/f_{B2}))$$

$$|A| = |A_{01}A_{02}|/(\sqrt{1 + (f/f_{B1})^2}\sqrt{1 + (f/f_{B2})^2})$$

$$|A|_{-3\text{dB}} = |A|/\sqrt{2} = |A_{01}A_{02}|/(\sqrt{1 + (f/f_{B1})^2}\sqrt{1 + (f/f_{B2})^2})$$

$$\sqrt{2} = \sqrt{1 + (f/f_{B1})^2}\sqrt{1 + (f/f_{B2})^2}$$

$$2 = (1 + (f/f_{B1})^2)(1 + (f/f_{B2})^2)$$

$$2 = 1 + f^2/f_{B1}^2 + f^2/f_{B2}^2 + f^4/(f_{B1}f_{B2})^2$$

$$0 = -1 + f^2/f_{B1}^2 + f^2/f_{B2}^2 + f^4/(f_{B1}f_{B2})^2$$

$$f_{B1} = 1.3636\text{MHz}$$

$$f_{B2} = 1.5\text{MHz}$$

$$f_b = 918.998\text{kHz}$$

(c) If $A_{01} = -50 \text{ V/V}$, what is the bandwidth of each subcircuit and the full circuit?

$$A_{01}A_{02} = -100 ; A_{01} = -50, A_{02} = 2$$

$$\beta_1 = 1/(1 - A_{01}) = 1/51$$

$$f_{B1} = f_t / 51 = 15\text{MHz}/51 = 294.118\text{kHz}$$

$$\beta_2 = 1/A_{02} = 1/2$$

$$f_{B2} = f_t/2 = 7.5\text{MHz}$$

$$0 = -1 + f^2/f_{B1}^2 + f^2/f_{B2}^2 + f^4/(f_{B1}f_{B2})^2$$

$$f_b = 293.667\text{kHz}$$

P2)

An inverting amplifier circuit is configured to provide a DC gain of $A_0 = -5$ V/V using an op amp with a gain bandwidth product of 1 MHz.

(a) Find the minimum slew rate required for the op amp such that the amplifier circuit is ensured to be small signal bandwidth limited; that is, slew rate limiting does not reduce the effective bandwidth of the circuit below f_B . Assume the amplifier is powered by ± 5 V supplies and is rail-to-rail.

$$A_0 = -5$$

$$f_t = 1\text{M}$$

$$A_0 = -R_1/R_2$$

$$|B| = R_2/(R_2+R_1) = 1/(1+R_1/R_2) = 1/(1-A_0) = 1/6$$

$$f_B = B f_t = 1\text{MHz}/6 = 166.666\text{kHz}$$

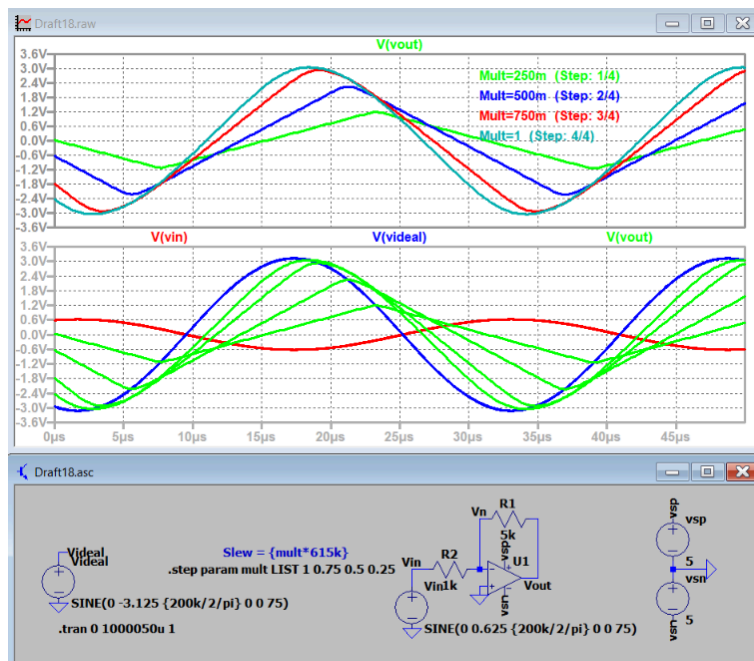
$$f_B \cdot 2\pi \cdot 5\text{V} = 5.236\text{ MV/s} = 5.236\text{ V/us}$$

(b) Assuming an input signal of $v_i = 0.625 \sin(2 \times 10^5 t + 75^\circ)$ V, what is the minimum slew rate to prevent slew rate limiting of the output?

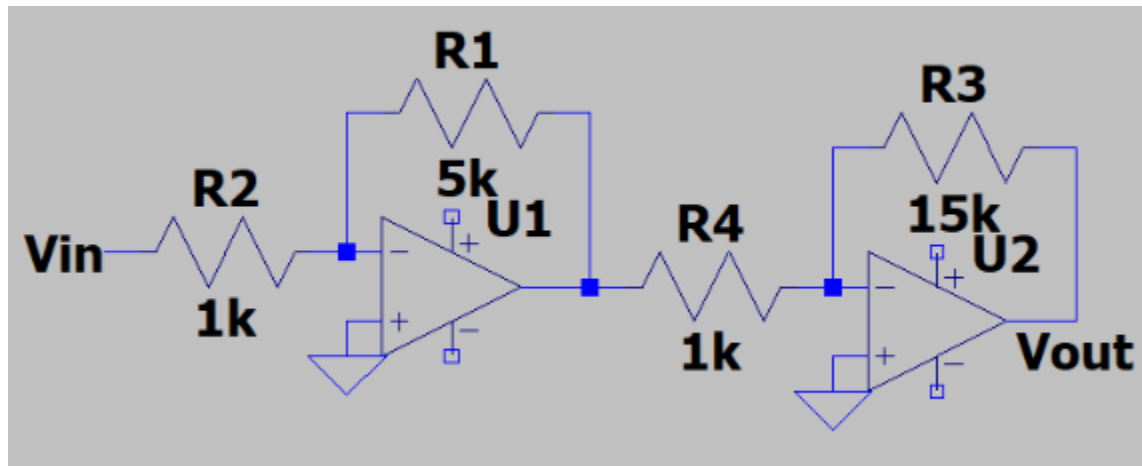
$$v_o = 3.125 \sin(200k t) \text{ V}$$

$$3.125 \cdot 200k = 615k\text{V/s} = 0.615\text{V/us}$$

(c) Simulate the circuit in LTspice with the voltage input signal from (b). Pick realistic values for the resistors to produce the given gain. For the op amp: use UniversalOpAmp2, which is a simplistic op amp model which includes slew rate. You'll need to edit the amplifier (right-click) to modify the GBP (GBW in LTspice) and slew rate (unit: V/s). Evaluate the simulation using the slew rate calculated for (b) and at values 25%, 50%, and 75% of this value. For each, plot the output voltage against the ideal output voltage. To get the ideal output voltage: either modify the trace of the input voltage (multiply by the gain) or produce another voltage source that outputs the ideal output.



(d) It is desired to use the inverting amplifier circuit above (unmodified) as a component of a larger circuit to provide an effective gain of $A_0 = 75 \text{ V/V}$. Design a cascaded circuit to produce this gain using only resistors and the op amp as given above (use as many as needed). What is the effective bandwidth and closed-loop gain bandwidth product of the resulting circuit?



$$A_0 = A_{01} \cdot A_{02} ; A_{02} = A_0 / A_{01} = 75 / -5 = -15$$

$$|B_1| = R_2 / (R_2 + R_1) = 1/6$$

$$|B_2| = R_4 / (R_4 + R_3) = 1/16$$

$$f_{B1} = |B_1| f_t = 1 \text{ MHz} / 6 = 166.666 \text{ kHz}$$

$$f_{B2} = |B_2| f_t = 1 \text{ MHz} / 16 = 62.5 \text{ kHz}$$

$$A = A_{01} / (1 + (f/f_{B1})) \cdot A_{02} / (1 + (f/f_{B2})) = A_{01} \cdot A_{02} / (1 + (f/f_{B1}))(1 + (f/f_{B2}))$$

$$|A| = |A_{01}| \cdot |A_{02}| / \sqrt{1 + (f^2/f_{B1}^2)} \sqrt{1 + (f^2/f_{B2}^2)}$$

$$|A|_{-3\text{dB}} = |A_{01}| / \sqrt{2} = |A_{01}| \cdot |A_{02}| / \sqrt{1 + (f^2/f_{B1}^2)} \sqrt{1 + (f^2/f_{B2}^2)}$$

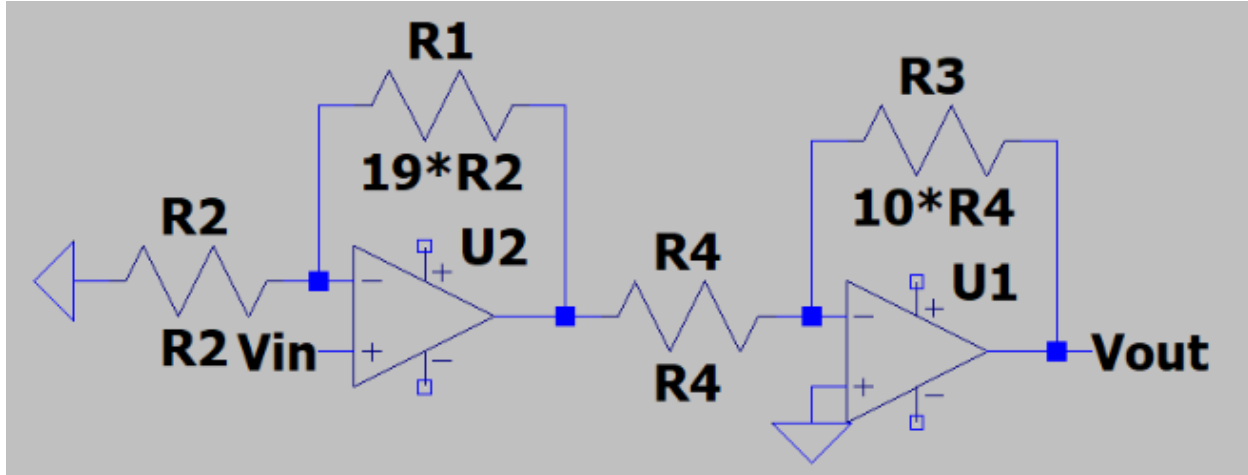
$$2 = (1 + (f^2/f_{B1}^2)) (1 + (f^2/f_{B2}^2)) = 1 + f^2/f_{B1}^2 + f^2/f_{B2}^2 + f^4/(f_{B1}f_{B2})^2$$

$$f_b = 55.838 \text{ kHz}$$

$$f_t = f_b \cdot A_0 = 55.838 \text{ kHz} \cdot 75 = 4.188 \text{ MHz}$$

P3) Book Problem 6.35.

A cascaded amplifier consists of an op amp OA1, operating as a noninverting amplifier with $A_0 = +20$ V/V, followed by an op amp OA2, operating as an inverting amplifier with $A_0 = -10$ V/V. Sketch the circuit; then find the minimum values of f_{t1} , SR_1 , f_{t2} , and SR_2 needed to ensure an overall bandwidth of 100 kHz with a full-power output signal of 5 V (rms).



$$A_{01} = 20$$

$$A_{02} = -10$$

$$|B_1| = 1/20$$

$$|B_2| = 1/11$$

$$f_{t1} = f_{B1} \cdot 20$$

$$f_{t2} = f_{B2} \cdot 11$$

$$|A|_{-3dB} = |A_{01}| / \sqrt{2} = |A_{01}| \cdot |A_{02}| / \sqrt{(1 + (f^2 / f_{B1}^2)) \sqrt{1 + (f^2 / f_{B2}^2)}}$$

$$2 = (1 + (f^2 / f_{B1}^2)) \cdot (1 + (f^2 / f_{B2}^2))$$

$$2 = (1 + (100\text{kHz}^2 / f_{B1}^2)) \cdot (1 + (100\text{kHz}^2 / f_{B2}^2))$$

one equation two unknowns, assume $f_{B1} = f_{B2}$ to minimize f_t

$$2 = (1 + (100\text{kHz}^2 / f_{B2}^2))^2$$

$$\sqrt{2} - 1 = 100\text{kHz}^2 / f_{B2}^2$$

$$f_{B2} = 100\text{kHz} / \sqrt{\sqrt{2} - 1} = 155.377\text{kHz}$$

$$f_{t1} = f_{B1} \cdot 20 = 3.108\text{MHz}$$

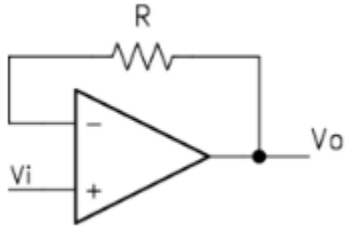
$$f_{t2} = f_{B2} \cdot 11 = 1.709\text{MHz}$$

$$SR_2 = 100\text{k} \cdot 2\pi \cdot 5 \cdot \sqrt{2} = 4.443 \text{ V/us}$$

$$SR_1 = 100\text{k} \cdot 2\pi \cdot 5 / 10 \cdot \sqrt{2} = 0.444 \text{ V/us}$$

P4)

Show that a voltage follower based on a current feedback amplifier with feedback resistor $R = 10 \text{ k}\Omega$ (shown below), does indeed provide an ideal gain of $A_{\text{ideal}} = 1$. Further, with the internal components of $z(jf)$ known as $R_{\text{eq}} = 500 \text{ k}\Omega$ and $C_{\text{eq}} = 2 \text{ pF}$, what is the gain error of the circuit at $f = 0 \text{ Hz}$, 100 kHz , 1 MHz , and 10 MHz ?



$$I_n = (V_i - V_o)/R$$

$$V_o = I_n \cdot Z$$

$$V_o/Z = (V_i - V_o)/R = V_i/R - V_o/R$$

$$V_o/Z + V_o/R = V_i/R$$

$$R V_o + Z V_o = Z V_i$$

$$V_o/V_i = Z / (R + Z) = 1 / (1 + R/Z)$$

$$A = V_o/V_i \approx 1 \text{ as long as } R \ll Z$$

$$V_o/I_n = Z(jf) = R_{\text{eq}} \parallel C_{\text{eq}} = R_{\text{eq}} * 1/(j 2\pi f C) / (R_{\text{eq}} + 1/(j 2\pi f C))$$

$$Z(jf) = R_{\text{eq}} / ((j 2\pi f C)(R_{\text{eq}} + 1/(j 2\pi f C))) = R_{\text{eq}} / (1 + (R_{\text{eq}} j 2\pi f C)) = R_{\text{eq}} / (1 + jf/f_b)$$

$$j f/f_b = (R_{\text{eq}} j 2\pi f C)$$

$$f_b = j f / (R_{\text{eq}} j 2\pi f C) = 1 / (R_{\text{eq}} 2\pi C) = 1 / (500 \text{ k} 2\pi 2\text{p}) = 159.155 \text{ kHz}$$

$$Z = R_{\text{eq}} / (1 + jf/f_b) = 500 \text{ k} / (1 + jf/159.155 \text{ kHz})$$

$$A = 1 / (1 + R/Z) = 1 / (1 + 10 \text{ k} / (500 \text{ k} / (1 + jf/159.155 \text{ kHz})))$$

$$GE = (A_{\text{ideal}} - A)/A_{\text{ideal}} = 1 - A$$

$f=0 \text{ Hz}$:

$$Z = 500 \text{ k} / (1 + 0/159.155 \text{ kHz}) = 500 \text{ k}$$

$$A = 1 / (1 + 10 \text{ k}/500 \text{ k}) = 0.98039$$

$$GE = 1.9607\%$$

$f=100 \text{ kHz}$:

$$Z = 500 \text{ k} / (1 + 0/159.155 \text{ kHz}) = 307.065 \text{ k}$$

$$A = 1 / (1 + 10 \text{ k}/500 \text{ k}) = 0.96846$$

$$GE = 3.1539\%$$

$f=1 \text{ MHz}$:

$$Z = 500 \text{ k} / (1 + 0/159.155 \text{ kHz}) = 68.651 \text{ k}$$

$$A = 1 / (1 + 10 \text{ k}/500 \text{ k}) = 0.87286$$

$$GE = 12.7144\%$$

$f=10 \text{ MHz}$:

$$Z = 500 \text{ k} / (1 + 0/159.155 \text{ kHz}) = 7.833 \text{ k}$$

$$A = 1 / (1 + 10 \text{ k}/500 \text{ k}) = 0.43924$$

$$GE = 56.0756\%$$